

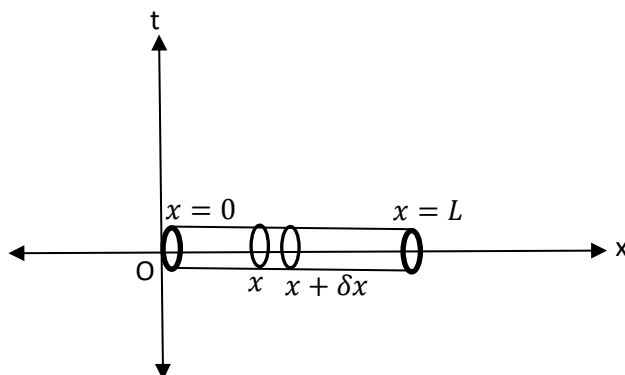
Unit-10

Separation of Variables

One Dimensional Heat Equation

Derivation of the Equation

Consider a homogeneous rod of length L conducting heat. Let this rod be placed along x –axis with one end at the origin ($x = 0$) and the other end at a point $x = L$.



We now make the following assumptions:

1. The rod has constant density ρ
2. The rod has uniform cross-section A
3. The lateral surface of the rod is insulated so that there is no heat gain or loss through it
4. Heat flows only in the x -direction
5. The rod is sufficiently thin so that the temperature at all points of any cross-section is same.

Let $u(x,t)$ be the temperature of the rod at any point $x \in [0, L]$ at any time t . Then, the amount of heat in a small segment of the rod between x and $x + \delta x$ is given by

$$q(x, t) = \int_x^{x+\delta x} c\rho A u(y, t) dy \quad \dots \quad (1),$$

where, c is the specific heat of the rod.

Now, by Fourier law of heat conduction, the rate of propagation of heat is

$$-\kappa A \frac{\partial u}{\partial x},$$

where, κ is the thermal conductivity of the medium.

Since the rate of change of heat content in the segment of the rod between x and $x + \delta x$ is equal to the net heat flow into the segment of the rod, we have

$$\frac{\partial q}{\partial t} = \kappa A \left[\left(\frac{\partial u}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial u}{\partial x} \right)_x \right] \quad \dots \quad (2)$$

Using mean value theorem of integral calculus in (1), we get

$$\begin{aligned} \frac{\partial q}{\partial t} &= \frac{\partial}{\partial t} \int_x^{x+\delta x} c\rho A u(y, t) dy \\ &= c\rho A \int_x^{x+\delta x} \frac{\partial u}{\partial t} dy \end{aligned}$$

$$\Rightarrow \frac{\partial q}{\partial t} = c\rho A \left(\frac{\partial u}{\partial t} \right)_{y=\xi} \delta x, \quad \xi \in [x, x + \delta x] \quad \dots \quad (3)$$

From (2) and (3), we have

$$c\rho A \left(\frac{\partial u}{\partial t} \right)_{y=\xi} \delta x = \kappa A \left[\left(\frac{\partial u}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial u}{\partial x} \right)_x \right]$$

Taking limit as $\delta x \rightarrow 0$, we get

$$\begin{aligned} c\rho A \frac{\partial u}{\partial t} &= \kappa A \frac{\partial^2 u}{\partial x^2} \\ \Rightarrow \frac{\partial u}{\partial t} &= \frac{\kappa}{c\rho} \frac{\partial^2 u}{\partial x^2} \end{aligned}$$

$$\Rightarrow u_t = \alpha^2 u_{xx} \quad \dots \quad (4),$$

where, $\alpha^2 = \frac{\kappa}{c\rho}$ is the thermal diffusivity, $u_t = \frac{\partial u}{\partial t}$ and $u_{xx} = \frac{\partial^2 u}{\partial x^2}$.

Equation (4) is the required one-dimensional heat conduction equation.

Initial and Boundary Conditions

Let us suppose that the initial temperature in the rod be given by

$$u(x, 0) = f(x), \quad 0 \leq x \leq L.$$

Also, we assume that the rod has fixed temperature at the ends of the rod $x = 0$ and $x = L$. Thus,

$$u(0, t) = 0, u(L, t) = 0, \quad t > 0.$$

Hence, the one-dimensional heat equation is

$$PDE: \quad u_t = \alpha^2 u_{xx}$$

together with initial condition

$$IC: \quad u(x, 0) = f(x), \quad 0 \leq x \leq L$$

and boundary conditions

$$BC: \quad u(0, t) = 0, u(L, t) = 0, \quad t > 0.$$

Variable Separation Solution of the One-Dimensional Heat Equation

One dimensional heat equation is

$$u_t = \alpha^2 u_{xx}, \quad \text{where } \alpha^2 = \frac{\kappa}{c\rho} \quad \dots \quad (1)$$

together with the initial condition

$$u(x, 0) = f(x), \quad 0 \leq x \leq L$$

and the boundary conditions

$$u(0, t) = 0, u(L, t) = 0, \quad t > 0 \quad \dots \quad (2).$$

Let

$$u(x, t) = X(x)T(t) \quad \dots \quad (3)$$

Be the variable separable solution of (1). Then,

$$u_t = \frac{\partial u}{\partial t} = X\dot{T} \quad \text{and} \quad u_{xx} = \frac{\partial^2 u}{\partial x^2} = X''T,$$

where, the dot (.) denote the derivative with respect to t and prime (') denote the derivative with respect to x.

Then, equation (1) reduces to

$$X\dot{T} = \alpha^2 X''T$$

$$\text{or, } \frac{X''}{X} = \frac{1}{\alpha^2} \frac{\dot{T}}{T} = -\lambda, \quad \text{say} \quad \dots \quad (4)$$

This implies

$$X'' + \lambda X = 0 \quad \dots \quad (5)$$

and

$$\dot{T} + \alpha^2 \lambda T = 0 \quad \dots \quad (6)$$

From Equation (5)

The A.E. is

$$m^2 + \lambda = 0 \Rightarrow m = \pm i\sqrt{\lambda}$$

Thus, the general solution of (5) is

$$X = c_1 \cos \sqrt{\lambda}x + c_2 \sin \sqrt{\lambda}x \quad \dots \quad (7)$$

Using the boundary conditions $u(0, t) = 0, u(L, t) = 0$, we get

$$X(0)T(t) = 0 \quad \text{and} \quad X(L)T(t) = 0$$

Since $T(t) \neq 0$, we have

$$X(0) = 0 \quad \text{and} \quad X(L) = 0$$

Using $X(0) = 0$ in (7), we get

$$0 = c_1 \cdot 1 + c_2 \cdot 0 \Rightarrow c_1 = 0.$$

Again, using $X(L) = 0$ in (7), we get

$$\begin{aligned} 0 &= 0 + c_2 \sin \sqrt{\lambda}L \\ \Rightarrow c_2 \sin \sqrt{\lambda}L &= 0 \\ \Rightarrow \sin \sqrt{\lambda}L &= 0 \Rightarrow \sqrt{\lambda}L = n\pi \\ \Rightarrow \sqrt{\lambda} &= \frac{n\pi}{L} \Rightarrow \lambda = \frac{n^2\pi^2}{L^2} \end{aligned}$$

Hence, the general solution of (7) becomes

$$X = c_2 \sin \frac{n\pi}{L}, \quad n \in Z^+$$

Setting $c_2 = 1 \neq 0$, in particular, the eigen functions associated with the eigen values are

$$X_n(x) = \sin \frac{n\pi x}{L}, \quad n \in Z^+, \text{ where } \frac{n\pi}{L} = \sqrt{\lambda}.$$

Next, from equation (6), we have

$$\begin{aligned} \dot{T} + \alpha^2 \lambda T &= 0 \Rightarrow \dot{T} + \alpha^2 \frac{n^2\pi^2}{L^2} T = 0 \\ \Rightarrow \dot{T} &= -\alpha^2 \frac{n^2\pi^2}{L^2} T \Rightarrow \frac{\dot{T}}{T} = -\alpha^2 \frac{n^2\pi^2}{L^2} \\ \Rightarrow \frac{1}{T} \frac{dT}{dt} &= -\alpha^2 \frac{n^2\pi^2}{L^2} \Rightarrow \frac{1}{T} dT = -\alpha^2 \frac{n^2\pi^2}{L^2} dt \end{aligned}$$

Integration implies

$$\begin{aligned} \int \frac{1}{T} dT &= -\alpha^2 \frac{n^2\pi^2}{L^2} \int dt \\ \Rightarrow \ln T &= -\alpha^2 \frac{n^2\pi^2}{L^2} t, \text{ neglecting arbitrary constant} \end{aligned}$$

$$\Rightarrow T = e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t}$$

$$\Rightarrow T_n(t) = e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t}, n \in \mathbb{Z}^+$$

Therefore, the function

$$u_n(x, t) = X_n(x)T_n(t)$$

$$= e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \sin \frac{n\pi x}{L}$$

$$u_n(x, t) = e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \sin \frac{n\pi x}{L}$$

Are solutions of the one-dimensional heat equation (1) under the given boundary conditions.

These solutions are also called the fundamental solutions of the heat equation.

It now remains to satisfy the initial condition $u(x, 0) = f(x), 0 \leq x \leq L$.

Consider

$$u(x, t) = \sum_{n=1}^{\infty} b_n u_n(x, t)$$

As any linear combination of the solutions. Then,

$$u(x, t) = \sum_{n=1}^{\infty} b_n e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \sin \frac{n\pi x}{L} \quad \dots \quad (8)$$

Using the initial condition $u(x, 0) = f(x)$, assuming the series on RHS converges and satisfy (1) along with the boundary conditions in equation (8), we get

$$\sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} = f(x)$$

Which is a Fourier sine series of $f(x)$ with period $2L$, so that the series on LHS converges to the initial temperature $f(x)$. Then,

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

Hence, the formal solution of the one-dimensional heat equation (1) with the given boundary conditions and the initial condition is

$$u(x, t) = \sum_{n=1}^{\infty} b_n e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \sin \frac{n\pi x}{L}, \text{ where } b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx, n \in \mathbb{Z}^+.$$

Ex-10.1

- Find the ordinary differential equations to represent the following partial differential equation:

b) $u_{xx} + u_{yy} + xu = 0$

Solution

The given PDE is

$$u_{xx} + u_{yy} + xu = 0 \quad \dots \quad (1)$$

Let

$$u(x, y) = X(x)Y(y) \quad \dots \quad (2)$$

be the variable separable solution of (1).

Then, from (2), we have

$$u_x = X'Y, u_{xx} = X''Y$$

and

$$u_y = XY', u_{yy} = XY''.$$

Using these values, equation (1) reduces to

$$\begin{aligned} X''Y + XY'' + xXY &= 0 \\ \Rightarrow X''Y + xXY &= -XY'' \\ \Rightarrow (X'' + xX)Y &= -XY'' \\ \Rightarrow \frac{(X'' + xX)}{X} &= -\frac{Y''}{Y} \\ \Rightarrow \frac{X''}{X} + x &= -\frac{Y''}{Y} \\ \Rightarrow \frac{X''}{X} + x &= -\frac{Y''}{Y} = -\lambda, \text{ say} \end{aligned}$$

Therefore,

$$\frac{X''}{X} + x = -\lambda \quad \dots \quad (3)$$

$$-\frac{Y''}{Y} = -\lambda \quad \dots \quad (4)$$

From (3),

$$\begin{aligned} \frac{X''}{X} + x &= -\lambda \\ \Rightarrow \frac{X''}{X} &= -x - \lambda \\ \Rightarrow X'' &= -(x + \lambda)X \\ \Rightarrow X'' + (x + \lambda)X &= 0. \end{aligned}$$

And, from (4), we get

$$\begin{aligned} -\frac{Y''}{Y} &= -\lambda \\ \Rightarrow Y'' &= \lambda Y \\ \Rightarrow Y'' - \lambda Y &= 0. \end{aligned}$$

Therefore, the required ODE that represent the PDE (1) are:

$$X'' + (x + \lambda)X = 0; \quad Y'' - \lambda Y = 0.$$

2. Find the solution of the heat conduction equation:

$$\begin{aligned} u_{xx} &= u_t, \quad 0 \leq x \leq L, t > 0 \\ u(0, t) &= u(L, t) = 0 \\ u(x, 0) &= \begin{cases} L - x, & \text{for } 0 < x \leq \frac{L}{2} \\ x, & \frac{L}{2} \leq x < L \end{cases} \end{aligned}$$

Solution

The given heat conduction equation is

$$u_{xx} = u_t, \quad 0 \leq x \leq L, t > 0 \quad \dots \quad (1)$$

together with the boundary conditions

$$u(0, t) = u(L, t) = 0 \quad \dots \quad (2)$$

and the initial condition

$$u(x, 0) = \begin{cases} L - x, & \text{for } 0 < x \leq \frac{L}{2} \\ x, & \frac{L}{2} \leq x < L \end{cases} \quad \dots \quad (3)$$

Let

$$u(x, t) = X(x)T(t) \quad \dots \quad (4)$$

be the variable separation solution of (1).

Then,

$$u_t = X\dot{T}, \quad u_{xx} = X''T.$$

Equation (1) now reduces to

$$X''T = X\dot{T}$$

$$\frac{X''}{X} = \frac{\dot{T}}{T} = -\lambda, \text{ say}$$

Then, we get

$$\frac{X''}{X} = -\lambda \Rightarrow X'' + \lambda X = 0 \quad \dots \quad (5)$$

and

$$\frac{\dot{T}}{T} = -\lambda \Rightarrow \dot{T} + \lambda T = 0 \quad \dots \quad (6)$$

The A.E. of (5) is

$$m^2 + \lambda = 0 \Rightarrow m = \pm i\sqrt{\lambda}.$$

Therefore, the general solution of (5) is

$$X = c_1 \cos\sqrt{\lambda}x + c_2 \sin\sqrt{\lambda}x \quad \dots \quad (7)$$

Next, using the boundary conditions (2), we have

$$\begin{aligned} u(0, t) &= u(L, t) = 0 \\ \Rightarrow X(0)T(t) &= 0 \text{ and } X(L)T(t) = 0 \end{aligned}$$

Since $T(t) \neq 0$, we must have $X(0) = 0$ and $X(L) = 0$.

Using $X(0) = 0$, equation (7) gives

$$0 = c_1 \cdot 1 + 0 \Rightarrow c_1 = 0$$

And using $X(L) = 0$, (7) gives

$$0 = 0 + c_2 \sin \sqrt{\lambda} L, \quad \therefore c_1 = 0$$

$$\text{Or,} \quad c_2 \sin \sqrt{\lambda} L = 0$$

$$\text{Or,} \quad \sin \sqrt{\lambda} L = 0$$

$$\text{Or,} \quad \sqrt{\lambda} L = n\pi \Rightarrow \sqrt{\lambda} = \frac{n\pi}{L} \Rightarrow \lambda = \frac{n^2 \pi^2}{L^2}, n \in Z^+$$

Hence, (7) becomes

$$X = c_2 \sin \frac{n\pi}{L} x, n \in Z^+$$

Choosing $c_2 = 1 \neq 0$, in particular, we have

$$X_n(x) = \sin \frac{n\pi}{L} x, n \in Z^+.$$

Again from (6), we have

$$\dot{T} + \lambda T = 0$$

$$\Rightarrow \dot{T} + \frac{n^2 \pi^2}{L^2} T = 0$$

$$\Rightarrow \frac{\dot{T}}{T} = -\frac{n^2 \pi^2}{L^2}$$

$$\Rightarrow \frac{1}{T} \frac{dT}{dt} = -\frac{n^2 \pi^2}{L^2}$$

$$\Rightarrow \frac{1}{T} dT = -\frac{n^2 \pi^2}{L^2} dt$$

Integration implies

$$\int \frac{1}{T} dT = \int -\frac{n^2 \pi^2}{L^2} dt$$

$$\ln T = -\frac{n^2 \pi^2}{L^2} t, \quad \text{neglecting arbitrary constant.}$$

$$T = e^{-\frac{n^2 \pi^2}{L^2} t}, \quad n \in Z^+$$

Therefore, as a sequence

$$T_n(t) = e^{-\frac{n^2\pi^2}{L^2}t}, \quad n \in \mathbb{Z}^+$$

And, therefore, the eigen function becomes

$$u_n(x, t) = X_n(x)T_n(t)$$

$$u_n(x, t) = \sin \frac{n\pi}{L}x \cdot e^{-\frac{n^2\pi^2}{L^2}t}, \quad n \in \mathbb{Z}^+$$

$$u_n(x, t) = e^{-\frac{n^2\pi^2}{L^2}t} \sin \frac{n\pi}{L}x, \quad n \in \mathbb{Z}^+$$

Also, the initial condition is,

$$u(x, 0) = \begin{cases} L - x, & \text{for } 0 < x \leq \frac{L}{2} \\ x, & \frac{L}{2} \leq x < L \end{cases}$$

We now consider

$$u(x, t) = \sum_{n=1}^{\infty} b_n u_n(x, t)$$

$$\text{or,} \quad u(x, t) = \sum_{n=1}^{\infty} b_n e^{-\frac{n^2\pi^2}{L^2}t} \sin \frac{n\pi}{L}x \quad \dots \quad (8),$$

which is a Fourier sine series of $u(x, 0)$ with period $2L$.

Therefore,

$$\begin{aligned} b_n &= \frac{2}{L} \int_0^L u(x, 0) \sin \frac{n\pi x}{L} dx \\ &= \frac{2}{L} \left[\int_0^{\frac{L}{2}} (L - x) \sin \frac{n\pi x}{L} dx + \int_{\frac{L}{2}}^L x \sin \frac{n\pi x}{L} dx \right] \end{aligned}$$

We first consider $\int_0^{\frac{L}{2}} (L - x) \sin \frac{n\pi x}{L} dx$

$$\begin{aligned} \int_0^{\frac{L}{2}} (L - x) \sin \frac{n\pi x}{L} dx &= \left[-(L - x) \times \frac{L}{n\pi} \cos \frac{n\pi x}{L} \right]_0^{\frac{L}{2}} - \frac{L}{n\pi} \int_0^{\frac{L}{2}} \cos \frac{n\pi x}{L} dx \\ &= \left[-\left(L - \frac{L}{2}\right) \times \frac{L}{n\pi} \cos \frac{n\pi}{2} + (L - 0) \frac{L}{n\pi} \cos 0 \right] - \frac{L}{n\pi} \int_0^{\frac{L}{2}} \cos \frac{n\pi x}{L} dx \end{aligned}$$

$$= -\frac{L^2}{2n\pi} \cos \frac{n\pi}{2} + \frac{L^2}{n\pi} - \frac{L}{n\pi} \int_0^{\frac{L}{2}} \cos \frac{n\pi x}{L} dx.$$

Next, we consider $\int_{\frac{L}{2}}^L x \sin \frac{n\pi x}{L} dx$

$$\begin{aligned} \int_{\frac{L}{2}}^L x \sin \frac{n\pi x}{L} dx &= \left[-\frac{L}{n\pi} x \cos \frac{n\pi x}{L} \right]_{\frac{L}{2}}^L + \frac{L}{n\pi} \int_{\frac{L}{2}}^L \cos \frac{n\pi x}{L} dx \\ &= \left[-\frac{L}{n\pi} \cdot L \cos n\pi + \frac{L}{n\pi} \cdot \frac{L}{2} \cos \frac{n\pi}{2} \right] + \frac{L}{n\pi} \int_{\frac{L}{2}}^L \cos \frac{n\pi x}{L} dx \\ &= \left[-\frac{L^2}{n\pi} \cdot \cos n\pi + \frac{L^2}{2n\pi} \cos \frac{n\pi}{2} \right] + \frac{L}{n\pi} \int_{\frac{L}{2}}^L \cos \frac{n\pi x}{L} dx \end{aligned}$$

Therefore,

$$\begin{aligned} & \frac{2}{L} \left[\int_0^{\frac{L}{2}} (L-x) \sin \frac{n\pi x}{L} dx + \int_{\frac{L}{2}}^L x \sin \frac{n\pi x}{L} dx \right] \\ &= \frac{2}{L} \left[-\frac{L^2}{2n\pi} \cos \frac{n\pi}{2} + \frac{L^2}{n\pi} - \frac{L}{n\pi} \int_0^{\frac{L}{2}} \cos \frac{n\pi x}{L} dx + \left[-\frac{L^2}{n\pi} \cdot \cos n\pi + \frac{L^2}{2n\pi} \cos \frac{n\pi}{2} \right] \right. \\ & \quad \left. + \frac{L}{n\pi} \int_{\frac{L}{2}}^L \cos \frac{n\pi x}{L} dx \right] \\ &= \frac{2}{L} \left[-\frac{L}{n\pi} \int_0^{\frac{L}{2}} \cos \frac{n\pi x}{L} dx + \frac{L}{n\pi} \int_{\frac{L}{2}}^L \cos \frac{n\pi x}{L} dx \right] \\ &= \frac{2}{n\pi} \left[-\int_0^{\frac{L}{2}} \cos \frac{n\pi x}{L} dx + \int_{\frac{L}{2}}^L \cos \frac{n\pi x}{L} dx \right] \\ &= \frac{2}{n\pi} \left[-\left[\frac{L}{n\pi} \sin \frac{n\pi x}{L} \right]_0^{\frac{L}{2}} + \left[\frac{L}{n\pi} \sin \frac{n\pi x}{L} \right]_{\frac{L}{2}}^L \right] \\ &= \frac{2L}{n^2\pi^2} \left[-\sin \frac{n\pi}{2} + 0 + \sin n\pi - \sin \frac{n\pi}{2} \right] \\ &= \frac{2L}{n^2\pi^2} \left[-2\sin \frac{n\pi}{2} \right] = \frac{-4L}{n^2\pi^2} \sin \frac{n\pi}{2} \end{aligned}$$

Therefore,

$$b_n = \frac{-4L}{n^2\pi^2} \sin \frac{n\pi}{2}$$

Hence, the required solution of the given heat conduction equation is

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} \frac{-4L}{n^2\pi^2} \sin \frac{n\pi}{2} \cdot e^{-\frac{n^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x \\ &= \frac{-4L}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \cdot e^{-\frac{n^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x \end{aligned}$$

Replacing n by $2n + 1$, we get

$$\begin{aligned} u(x, t) &= \frac{-4L}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^2} \sin \frac{(2n+1)\pi}{2} \cdot e^{-\frac{(2n+1)^2\pi^2}{L^2}t} \sin \frac{(2n+1)\pi}{L} x \\ &= \frac{-4L}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^2} \cdot 1 \cdot e^{-\frac{(2n+1)^2\pi^2}{L^2}t} \sin \frac{(2n+1)\pi}{L} x \\ \therefore u(x, t) &= \frac{-4L}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^2} e^{-\frac{(2n+1)^2\pi^2}{L^2}t} \sin \frac{(2n+1)\pi}{L} x \end{aligned}$$

3. Find the solution of the heat equation

$$u_{xx} = u_t, 0 < x < 40, t > 0$$

With boundary conditions

$$u(0, t) = 0 = u(40, t)$$

And with the initial condition

$$u(x, 0) = \begin{cases} 0, & \text{if } 0 \leq x < 10 \\ 50, & \text{if } 10 \leq x \leq 30 \\ 0, & \text{if } 30 \leq x \leq 40 \end{cases}$$

Solution

The given heat equation is

$$u_{xx} = u_t, 0 < x < 40, t > 0 \quad \dots \quad (1)$$

Together with the boundary conditions

$$u(0, t) = 0 = u(40, t) \quad \dots \quad (2)$$

And the initial condition

$$u(x, 0) = \begin{cases} 0, & \text{if } 0 \leq x < 10 \\ 50, & \text{if } 10 \leq x \leq 30 \\ 0, & \text{if } 30 \leq x \leq 40 \end{cases} \quad \dots \quad (3)$$

Let the variable separable solution of (1) be

$$u(x, t) = X(x)T(t) \quad \dots \quad (4)$$

Therefore,

$$u_t = X\dot{T}, \text{ and } u_{xx} = X''T$$

Then, equation (1) reduces to

$$X''T = X\dot{T}$$

$$\Rightarrow \frac{X''}{X} = \frac{\dot{T}}{T} = -\lambda, \text{ say}$$

Therefore, we get

$$X'' + \lambda X = 0 \quad \dots \quad (5)$$

$$\dot{T} + \lambda T = 0 \quad \dots \quad (6)$$

The A.E. of (5) is

$$m^2 + \lambda = 0 \quad \therefore m = \pm i\sqrt{\lambda}$$

So, the general solution of (1) is

$$X = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x \quad \dots \quad (7)$$

Now, the boundary conditions

$$u(0, t) = 0 \quad \text{and} \quad u(40, t) = 0$$

$$\Rightarrow X(0)T(t) = 0 \quad \text{and} \quad X(40)T(t) = 0$$

Since $T(t) \neq 0$, we get

$$X(0) = 0 \quad \text{and} \quad X(40) = 0$$

Using $X(0) = 0$ in (7), we get

$$0 = c_1 \cdot 1 + c_2 \cdot 0 \Rightarrow c_1 = 0$$

Again, using $X(40) = 0$ and $c_1 = 0$ in (7), we get

$$0 = 0 + c_2 \sin \sqrt{\lambda} 40 \Rightarrow c_2 \sin \sqrt{\lambda} 40 = 0$$

$$\Rightarrow \sin \sqrt{\lambda} 40 = 0 \Rightarrow \sqrt{\lambda} 40 = n\pi$$

$$\Rightarrow \sqrt{\lambda} = \frac{n\pi}{40} \Rightarrow \lambda = \frac{n^2 \pi^2}{1600}, \quad n \in Z^+$$

Thus, (7) becomes

$$X = c_2 \sin \frac{n\pi}{40} x$$

For particular, setting $c_2 = 1 \neq 0$, we get

$$X_n(x) = \sin \frac{n\pi}{40} x$$

Again, from (6)

$$\dot{T} + \lambda T = 0 \Rightarrow \frac{dT}{dt} = -\frac{n^2 \pi^2}{1600} T$$

$$\Rightarrow \frac{dT}{T} = -\frac{n^2 \pi^2}{1600} dt$$

Integration implies

$$\int \frac{dT}{T} = -\frac{n^2 \pi^2}{1600} \int dt$$

$$\Rightarrow \ln T = -\frac{n^2 \pi^2}{1600} t, \quad \text{neglecting integration constant.}$$

$$\Rightarrow T = e^{-\frac{n^2 \pi^2}{1600} t}$$

Therefore, as a sequence

$$T_n(t) = e^{-\frac{n^2 \pi^2}{1600} t}, \quad n \in Z^+$$

Therefore, the function

$$u_n(x, t) = X_n(x)T_n(t) \text{ becomes}$$

$$u_n(x, t) = \sin \frac{n\pi}{40} x e^{-\frac{n^2\pi^2}{1600}t}, \quad n \in Z^+$$

or, $u_n(x, t) = e^{-\frac{n^2\pi^2}{1600}t} \sin \frac{n\pi}{40} x, \quad n \in Z^+$

We now consider

$$u(x, t) = \sum_{n=1}^{\infty} b_n u_n(x, t)$$

That is,

$$u(x, t) = \sum_{n=1}^{\infty} b_n e^{-\frac{n^2\pi^2}{1600}t} \sin \frac{n\pi}{40} x, \quad n \in Z^+ \quad \dots \quad (8)$$

Using the initial condition (3) in (8), we have

$$\sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{40} x = f(x),$$

which is a Fourier sine series of $u(x, 0)$ with period $2L = 2 \times 40 = 80$. Then,

$$\begin{aligned} b_n &= \frac{2}{40} \int_0^{40} u(x, 0) \sin \frac{n\pi}{40} x dx \\ &= \frac{1}{20} \left[\int_0^{10} 0 \cdot \sin \frac{n\pi}{40} x dx + \int_{10}^{30} 50 \cdot \sin \frac{n\pi}{40} x dx + \int_{30}^{40} 0 \cdot \sin \frac{n\pi}{40} x dx \right] \\ &= \frac{1}{20} \left[0 + \int_{10}^{30} 50 \cdot \sin \frac{n\pi}{40} x dx + 0 \right] = \frac{50}{20} \left[\int_{10}^{30} \sin \frac{n\pi}{40} x dx \right] \\ &= \frac{5}{2} \frac{40}{n\pi} \left[-\cos \frac{n\pi}{40} x \right]_{10}^{30} = \frac{100}{n\pi} \left[-\cos \frac{n\pi}{40} \cdot 30 + \cos \frac{n\pi}{40} \cdot 10 \right] \\ &= \frac{100}{n\pi} \left[-\cos \frac{3n\pi}{4} + \cos \frac{n\pi}{4} \right] \end{aligned}$$

Therefore,

$$b_n = \frac{100}{n\pi} \left[\cos \frac{n\pi}{4} - \cos \frac{3n\pi}{4} \right]$$

Hence, the required solution of the given heat conduction equation is

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} \frac{100}{n\pi} \left[\cos \frac{n\pi}{4} - \cos \frac{3n\pi}{4} \right] e^{-\frac{n^2\pi^2}{1600}t} \sin \frac{n\pi}{40} x, \quad n \in Z^+ \\ &= \frac{100}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left[\cos \frac{n\pi}{4} - \cos \frac{3n\pi}{4} \right] e^{-\frac{n^2\pi^2}{1600}t} \sin \frac{n\pi}{40} x, \quad n \in Z^+ \end{aligned}$$

Solution of One-dimensional Heat Equation with Non-homogeneous Boundary Conditions

Here, we assume that the temperature at one end of the rod is kept constant to be T_2 and at the other end it kept constant to be T_1 . Then, the non-homogeneous boundary conditions are

$$u(0, t) = T_1, \quad u(L, t) = T_2, \quad t > 0.$$

The heat conduction equation is the same as before

$$u_t = \alpha^2 u_{xx} \quad 0 < x < L, \quad t > 0 \quad \dots \quad (1)$$

with the non-homogeneous boundary conditions

$$u(0, t) = T_1, \quad u(L, t) = T_2, \quad t > 0 \quad \dots \quad (2)$$

and the same initial condition

$$u(x, 0) = f(x), \quad 0 \leq x \leq L \quad \dots \quad (3).$$

We now wish to make the non-homogeneous conditions (2) homogeneous and solve.

Assume, at infinity ($t \rightarrow \infty$), that a steady temperature is acquired, say $v(x)$, which is independent of time t and the initial condition (3). Since $v(x)$ satisfies (1), we have

$$\begin{aligned} v_t &= \alpha^2 v_{xx} \\ \Rightarrow 0 &= \alpha^2 v_{xx}, \quad \text{since } v_t = 0 \\ \Rightarrow v_{xx} &= 0. \end{aligned}$$

Integration implies

$$v_x = c_1$$

Again, integrating we get

$$v = c_1 x + c_2 \quad \dots \quad (4).$$

This shows that the steady state temperature distribution $v(x)$ is a linear function of x . Also, $v(x)$ satisfies the boundary conditions

$$v(0) = T_1 \quad \text{and} \quad v(L) = T_2$$

which is valid for $T \rightarrow \infty$ as well.

Since $v(0) = T_1$, (4) reduces to

$$T_1 = c_1 \cdot 0 + c_2 \Rightarrow c_2 = T_1$$

Again, since $v(L) = T_2$ from (4) we get

$$T_2 = c_1 \cdot L + c_2 \Rightarrow T_2 = c_1 L + T_1 \Rightarrow c_1 = \frac{T_2 - T_1}{L}$$

Using these values, equation (4) becomes

$$v = \frac{T_2 - T_1}{L} x + T_1 \quad \dots \quad (5)$$

Consider

$$u(x, t) = v(x) + \omega(x, t) \quad \dots \quad (6)$$

where $v(x)$ is the steady state temperature distribution (independent of time t) and $\omega(x, t)$ is transient temperature distribution (time dependent). Then, from (1)

$$\begin{aligned} (v + \omega)_t &= \alpha^2 (v + \omega)_{xx} \\ \text{or, } v_t + \omega_t &= \alpha^2 v_{xx} + \alpha^2 \omega_{xx} \\ \text{or, } \omega_t &= \alpha^2 \omega_{xx}, \quad [\because v_t = 0 = v_{xx}] \end{aligned}$$

Also,

$$\omega(0, t) = u(0, t) - v(0) = T_1 - T_1 = 0 \quad \dots \quad (7)$$

$$\omega(L, t) = u(L, t) - v(L) = T_2 - T_2 = 0 \quad \dots \quad (8)$$

$$\omega(x, 0) = u(x, 0) - v(x) = f(x) - v(x) \quad \dots \quad (9)$$

From (7) and (8) we see that the boundary conditions for $\omega(x, t)$ is the same as that of the previous case. We now solve equations (7), (8) and (9) similarly as in the previous case to get

$$\omega(x, t) = \sum_{n=1}^{\infty} b_n e^{-\frac{n^2 \pi^2 \alpha^2}{L^2} t} \sin \frac{n \pi x}{L} \quad \dots \quad (10),$$

where,

$$b_n = \frac{2}{L} \int_0^L [f(x) - v(x)] \sin \frac{n \pi x}{L} dx, \quad n \in \mathbb{Z}^+ \quad \dots \quad (11)$$

Hence, from (5), (10) and (11), the general solution of (1) with the non-homogeneous boundary conditions (2) and the initial condition (3) is

$$u(x, t) = v(x) + \omega(x, t) = \left\{ \frac{T_2 - T_1}{L} x + T_1 \right\} + \sum_{n=1}^{\infty} b_n e^{-\frac{n^2 \pi^2 \alpha^2}{L^2} t} \sin \frac{n \pi x}{L},$$

where,

$$b_n = \frac{2}{L} \int_0^L \left[f(x) - \left\{ \frac{T_2 - T_1}{L} x + T_1 \right\} \right] \sin \frac{n \pi x}{L} dx, \quad n \in \mathbb{Z}^+.$$

Rod With Insulated Ends

In this section, we consider the rod of which both the ends are insulated so that there is heat flow through both the ends. Also, by Fourier law, the rate of flow of heat across any cross section is proportional to the rate of change of temperature in the direction.

So, the boundary conditions for no heat flow are:

$$u_x(0, t) = 0, \quad u_x(L, t) = 0, \quad t > 0$$

Now, the heat conduction equation becomes

$$u_t = \alpha^2 u_{xx}, \quad 0 < x < L, \quad t > 0 \quad \dots \quad (1)$$

with boundary conditions

$$u_x(0, t) = 0, \quad u_x(L, t) = 0, \quad t > 0, \quad [\text{no heat flow}] \quad \dots \quad (2)$$

and the initial condition

$$u(x, 0) = f(x), \quad 0 \leq x \leq L \quad \dots \quad (3)$$

Let

$$u(x, t) = X(x)T(t) \quad \dots \quad (4)$$

be the variable separable solution of the above system. Then,

$$u_t = X\dot{T}, \text{ and } u_{xx} = X''T.$$

Using these values in (4), we get

$$\begin{aligned} \alpha^2 X''T &= X\dot{T} \\ \Rightarrow \frac{X''}{X} &= \frac{1}{\alpha^2} \frac{\dot{T}}{T} = -\lambda, \text{ say.} \end{aligned}$$

Taking first and last ratios, we get

$$X'' + \lambda X = 0 \quad \dots \quad (5)$$

$$\dot{T} + \alpha^2 \lambda T = 0 \quad \dots \quad (6)$$

Now, the boundary conditions

$$\begin{aligned} u_x(0, t) = 0 \text{ and } u_x(L, t) = 0 &\Rightarrow X'(0)T(t) = 0 \text{ and } X'(L)T(t) = 0 \\ &\Rightarrow X'(0) = 0, X'(L) = 0 \text{ since } T(t) \neq 0 \end{aligned}$$

The A.E. of (5) is

$$m^2 + \lambda = 0 \quad (7)$$

We deal with the following two cases:

Case I [when $\lambda < 0$]

If $\lambda < 0$, then we may suppose $\lambda = -\mu^2$, where μ is real and positive. Then, from (7)

$$m^2 - \mu^2 = 0 \Rightarrow m^2 = \mu^2 \Rightarrow m = \pm \mu$$

Then, the general solution of (5) is

$$\begin{aligned} X(x) &= c_1 e^{\mu x} + c_2 e^{-\mu x} \\ \therefore X'(x) &= \mu c_1 e^{\mu x} - \mu c_2 e^{-\mu x} \end{aligned}$$

Now,

$$X'(0) = 0 \Rightarrow 0 = \mu c_1 \cdot 1 - \mu c_2 \cdot 1 \Rightarrow c_1 - c_2 = 0 \quad \dots \quad (8)$$

And

$$X'(L) = 0 \Rightarrow 0 = \mu c_1 e^{\mu L} - \mu c_2 e^{-\mu L} \Rightarrow c_1 e^{\mu L} - c_2 e^{-\mu L} = 0 \quad \dots \quad (9)$$

Solving (8) and (9), we get

$$c_1 = 0 \text{ and } c_2 = 0, \text{ the unique solution}$$

which is not possible and so, λ cannot be negative.

Case II [when $\lambda > 0$]

If $\lambda > 0$, then we may suppose $\lambda = \mu^2$, where μ is real and positive. Then, from (7)

$$m^2 + \mu^2 = 0 \Rightarrow m^2 = -\mu^2 \Rightarrow m = \pm i\mu$$

Then, the general solution of (5) is

$$X(x) = c_1 \sin \mu x + c_2 \cos \mu x \quad \dots \quad (10)$$

$$\therefore X'(x) = \mu c_1 \cos \mu x - \mu c_2 \sin \mu x$$

Now,

$$X'(0) = 0 \Rightarrow 0 = \mu c_1 \cdot 1 - \mu c_2 \cdot 0 \Rightarrow c_1 = 0$$

And

$$X'(L) = 0 \Rightarrow 0 = \mu c_1 \cos \mu L - \mu c_2 \sin \mu L \Rightarrow \mu c_2 \sin \mu L = 0, \text{ since } c_1 = 0$$

$$\mu c_2 \sin \mu L = 0 \Rightarrow \sin \mu L = 0 \Rightarrow \mu L = n\pi \Rightarrow \mu = \frac{n\pi}{L}, n \in Z^+$$

Therefore, $\lambda = \mu^2 = \frac{n^2 \pi^2}{L^2}, n \in Z^+$. Thus, (10) becomes

$$X(x) = c_2 \cos \frac{n\pi}{L} x$$

Choosing $c_2 = 1 \neq 0$, for particular, we get

$$X(x) = \cos \frac{n\pi}{L} x, \quad n \in Z^+$$

Also, from (6)

$$\begin{aligned} \dot{T} + \alpha^2 \lambda T &= 0 \\ \Rightarrow \dot{T} + \alpha^2 \frac{n^2 \pi^2}{L^2} T &= 0 \Rightarrow \dot{T} = -\alpha^2 \frac{n^2 \pi^2}{L^2} T \\ \Rightarrow \frac{dT}{dt} &= -\alpha^2 \frac{n^2 \pi^2}{L^2} T \Rightarrow \frac{dT}{T} = -\alpha^2 \frac{n^2 \pi^2}{L^2} dt \end{aligned}$$

Integration implies

$$\begin{aligned} \int \frac{dT}{T} &= -\alpha^2 \frac{n^2 \pi^2}{L^2} \int dt \\ \Rightarrow \ln T &= -\alpha^2 \frac{n^2 \pi^2}{L^2} t, \quad \text{neglecting integration constant} \\ \Rightarrow T &= e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \end{aligned}$$

Therefore,

$$T(t) = e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t}$$

Hence, the required fundamental solution of the heat conduction problem (1)-(3) is

$$\begin{aligned} u_0(x, t) &= 1, \\ u_n(x, t) &= e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \cos \frac{n\pi}{L} x, \quad n \in Z^+ \end{aligned}$$

To satisfy the initial condition (3), we assume $u(x, t)$ in the form

$$u(x, t) = \frac{b_0}{2} u_0(x, t) + \sum_{n=1}^{\infty} b_n u_n(x, t)$$

$$\therefore u(x, t) = \frac{b_0}{2} + \sum_{n=1}^{\infty} b_n e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \cos \frac{n\pi}{L} x,$$

which is Fourier cosine series of $f(x)$ with period $2L$ and

$$b_n = \frac{2}{L} \int \cos \frac{n\pi}{L} x f(x) dx, \quad n \in \mathbb{Z}^+.$$

Exercise-10.2

Find the steady state solution of the heat conduction equation $u_t = \alpha^2 u_{xx}$ if it satisfies the following set of boundary conditions

$$2. \quad u(0, t) = 30, u(40, t) = -10$$

Solution

The heat conduction equation is

$$u_t = \alpha^2 u_{xx} \quad \dots \quad (1)$$

and the assigned boundary conditions are

$$u(0, t) = 30, u(40, t) = -10 \quad \dots \quad (2)$$

Let the steady state temperature be $v(x)$. Then, it satisfies

$$v''(x) = 0$$

And the boundary conditions

$$v(0) = 30 \text{ and } v(40) = -10$$

Now,

$$v''(x) = 0 \Rightarrow v'(x) = c_1 \Rightarrow v(x) = c_1 x + c_2$$

Since $v(0) = 30$

$$v(x) = c_1 x + c_2 \Rightarrow v(0) = c_1 \cdot 0 + c_2 \Rightarrow 30 = c_2$$

$$\therefore c_2 = 30$$

Again, Since $v(40) = -10$

$$v(x) = c_1 x + c_2 \Rightarrow v(40) = c_1 \times 40 + c_2 \Rightarrow -10 = 40c_1 + 30 \Rightarrow 40c_1 = -40$$

$$\therefore c_1 = -1$$

Therefore, the steady state temperature is

$$v(x) = -x + 30$$